Karan Banerjee

Mechatronics Engineer

RESUME_1746

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Total Experience: 9 years

PROFESSIONAL SUMMARY

Mechatronics Engineer with 9 years of experience in product development, using CAD, SolidWorks, Arduino, ESP32. Experienced in electromechanical systems, integration tests and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: PLC, Sensors, SolidWorks, ESP32, CAD, Arduino, Control Systems, 3D Printing, Testing, Python, C++, Rapid Prototyping

Methods: Electromechanical systems, Integration tests, Product prototyping

PROFESSIONAL EXPERIENCE

Lead Mechatronics Engineer | DeltaPoint Innovations | Jan 2023 - Present

- Improved prototype throughput by 17% through actuator timing optimization
- Validated 773 integration test cases across mechanical and electrical interfaces
- Reduced assembly cost by 18% after redesigning electromechanical fixtures
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Mechatronics Engineer | Evergrid Technologies | Jan 2020 - Dec 2022

- Reduced assembly cost by 32% after redesigning electromechanical fixtures
- Raised measured actuator positioning consistency by 30% after calibration

Associate Mechatronics Engineer | SparrowCloud Labs | Jul 2017 - Dec 2019

- Improved prototype throughput by 41% through actuator timing optimization
- Validated 570 integration test cases across mechanical and electrical interfaces

PROJECTS

Smart Dispensing Unit | C++, CAD, Sensors, PLC

- Built controller firmware, mechanical enclosure and fault signals.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

Lab Instrument Positioner | Sensors, Arduino, PLC, C++

- Designed stepper motion stages and calibration routines.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

EDUCATION

B.Tech in Mechatronics Engineering, Shivanta College of Technology, 2017
Relevant Methods: Electromechanical systems, Integration tests, Product prototyping

CERTIFICATIONS

Sensor Interfacing (2024)